

MAT211: Calculus-II

Practice Sheet 01

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Problem 0.1. Which of the following subsets of $\mathbb{R}^n$ is open? closed? neither? Prove your answer.

1. $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y > 0 \right\} \subset \mathbb{R}^2$

2. $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y \geq 0 \right\} \subset \mathbb{R}^2$

3. $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y > x \right\} \subset \mathbb{R}^2$

4. $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \neq 0 \right\} \subset \mathbb{R}^2$

1. *Openness.* Let $(x_0, y_0) \in S_1$. Then $y_0 > 0$. Choose $r = \frac{y_0}{2} > 0$. For any (x, y) with $\|(x, y) - (x_0, y_0)\| < r$, we have $|y - y_0| < r$, hence $y > y_0 - r = \frac{y_0}{2} > 0$. Thus the open ball centered at (x_0, y_0) of radius r is contained in S_1 . Hence S_1 is open. $\square$

1. *Not closed.* The point $(0, 0)$ is a limit point of S_1 (take sequence $(0, \frac{1}{n}) \in S_1$ converging to $(0, 0)$), but $(0, 0) \notin S_1$. Therefore S_1 is not closed. $\square$

2. *Closedness.* The complement $S_2^c = \{(x, y) : y < 0\}$ is open (by the same argument as in part (1)). Hence S_2 is closed. $\square$

2. *Not open.* Take $(0, 0) \in S_2$. Any open ball around $(0, 0)$ contains points with $y < 0$ (e.g., $(0, -\frac{r}{2})$), which are not in S_2 . Hence S_2 is not open. $\square$

3. *Openness.* Define $f(x, y) = y - x$, which is continuous on $\mathbb{R}^2$. Then $S_3 = f^{-1}((0, \infty))$. Since $(0, \infty)$ is open in $\mathbb{R}$ and f is continuous, the preimage is open. Alternatively, a direct ε - δ argument: for any $(x_0, y_0) \in S_3$, let $d = y_0 - x_0 > 0$ and choose $r = \frac{d}{2}$. If $\|(x, y) - (x_0, y_0)\| < r$, then $|x - x_0| < r$ and $|y - y_0| < r$, so

$$y - x = (y - y_0) + (y_0 - x_0) + (x_0 - x) > -r + d - r = d - 2r = 0.$$

Thus the ball is contained in S_3 . $\square$

3. *Not closed.* The point $(0, 0)$ is a limit point of S_3 (take sequence $(0, \frac{1}{n}) \in S_3$ converging to $(0, 0)$), but $(0, 0) \notin S_3$ because $0 \not> 0$. Hence S_3 is not closed. $\square$

4. *Openness.* Define $g(x, y) = xy$, which is continuous. Then $S_4 = g^{-1}(\mathbb{R} \setminus \{0\})$. Since $\mathbb{R} \setminus \{0\}$ is open and g is continuous, S_4 is open. Directly: for any (x_0, y_0) with $x_0 \neq 0$ and $y_0 \neq 0$, let $r = \min\{|x_0|, |y_0|\} > 0$. If $\|(x, y) - (x_0, y_0)\| < r$, then $|x| \geq |x_0| - |x - x_0| > |x_0| - r \geq 0$, and similarly $|y| > 0$, so $xy \neq 0$. Hence the ball is contained in S_4 . $\square$

4. *Not closed.* The point $(0, 0)$ is a limit point of S_4 (take sequence $(\frac{1}{n}, \frac{1}{n}) \in S_4$ converging to $(0, 0)$), but $(0, 0) \notin S_4$. Also, points on the axes are limit points not in the set. Thus S_4 is not closed. $\square$

Problem 0.2. Identify $\mathcal{M}_{m \times n}$ the set of $m \times n$ matrices, with $\mathbb{R}^{mn}$ in the obvious way.

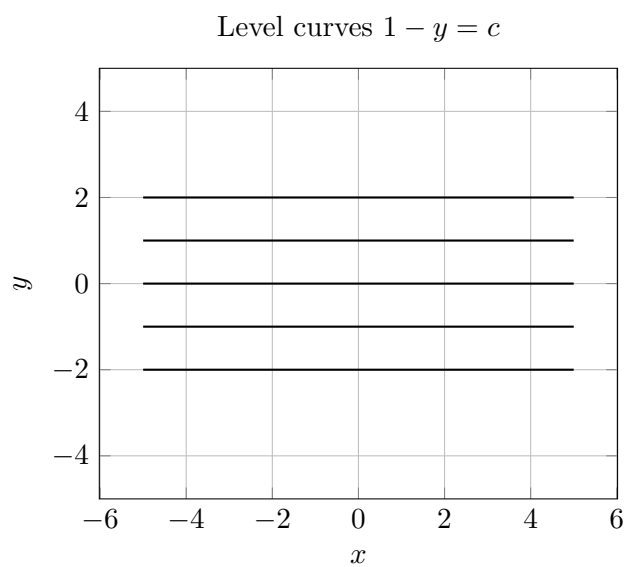
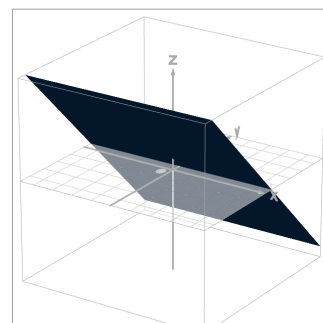
1. Prove that when $n = 2$ or 3 , the set of $n \times n$ matrices with nonzero determinant is an open subset of $\mathcal{M}_{n \times n}$.
2. Prove that the set of $n \times n$ matrices A satisfying $A^T A = I_n$ is a closed subset of $\mathcal{M}_{n \times n}$.

2(a). The determinant function $\det : \mathcal{M}_{n \times n} \rightarrow \mathbb{R}$ is a polynomial in the entries of a matrix, hence continuous with respect to the Euclidean topology on $\mathcal{M}_{n \times n} \cong \mathbb{R}^{n^2}$. The set of matrices with nonzero determinant is $\det^{-1}(\mathbb{R} \setminus \{0\})$. Since $\mathbb{R} \setminus \{0\}$ is open in $\mathbb{R}$ and $\det$ is continuous, the preimage is open. This argument holds for any n , in particular for $n = 2$ and $n = 3$. $\square$

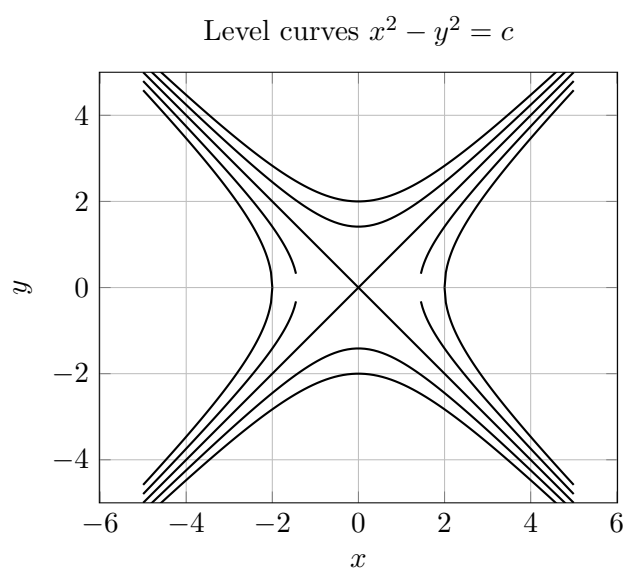
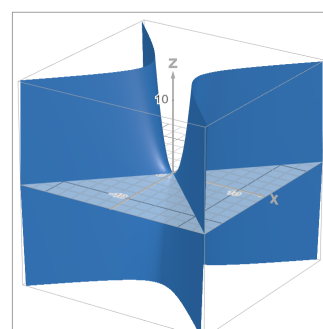
2(b). Define $F : \mathcal{M}_{n \times n} \rightarrow \mathcal{M}_{n \times n}$ by $F(A) = A^T A - I_n$. Each entry of $F(A)$ is a polynomial in the entries of A , so F is continuous (componentwise). The set $\{0\} \subset \mathcal{M}_{n \times n}$ is closed (singletons are closed in a metric space). Hence the preimage $F^{-1}(\{0\}) = \{A \in \mathcal{M}_{n \times n} : A^T A = I_n\}$ is closed. $\square$

Problem 0.3. Sketch the families of level curves and the graphs of the following functions f .

1. $f \begin{pmatrix} x \\ y \end{pmatrix} = 1 - y$
2. $f \begin{pmatrix} x \\ y \end{pmatrix} = x^2 - y^2$
3. $f \begin{pmatrix} x \\ y \end{pmatrix} = x^2 - y$
4. $f \begin{pmatrix} x \\ y \end{pmatrix} = xy$

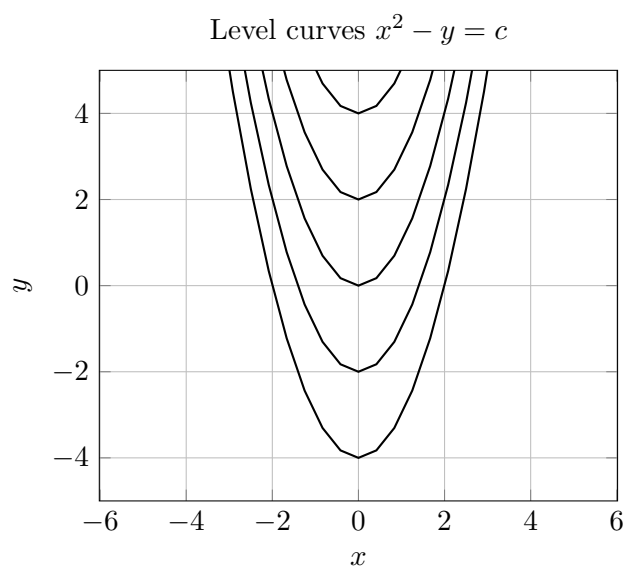
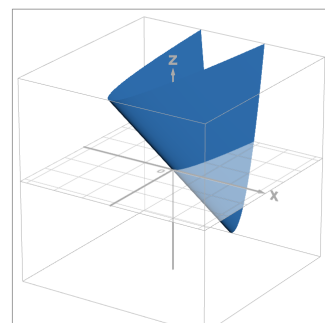
Figure 1: Horizontal lines $y = 1 - c$.

$$f(x, y) = 1 - y$$

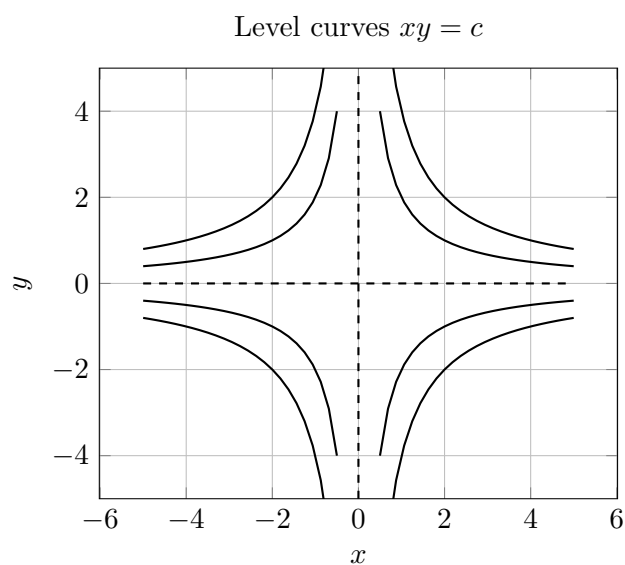
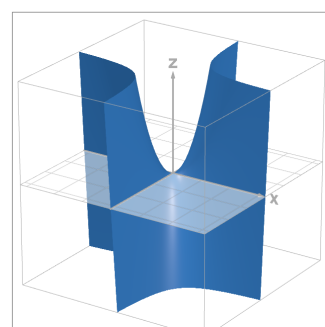
Figure 2: Graph of $f(x, y) = 1 - y$.Figure 3: Hyperbolas (lines for $c = 0$).

$$f(x, y) = x^2 - y^2$$

Figure 4: Graph of $f(x, y) = x^2 - y^2$.

Figure 5: Parabolas $y = x^2 - c$.

$$f(x,y) = x^2 - y$$

Figure 6: Graph of $f(x,y) = x^2 - y$.Figure 7: Rectangular hyperbolas (axes for $c = 0$).

$$f(x,y) = xy$$

Figure 8: Graph of $f(x,y) = xy$.

1. $f(x, y) = 1 - y$
2. $f(x, y) = x^2 - y^2$
3. $f(x, y) = x^2 - y$
4. $f(x, y) = xy$

Problem 0.4. Consider the surfaces

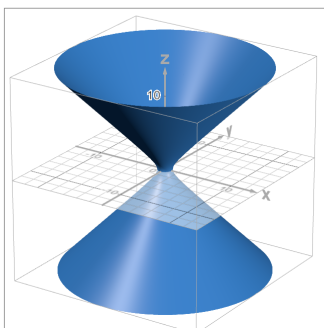
$$X = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 - z^2 = 1 \right\} \text{ and } Y = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 - z^2 = -1 \right\}$$

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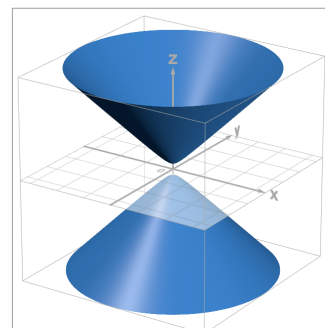
1. Sketch the surfaces.
2. Give a rigorous argument (not merely based on your pictures) that every pair of points of X can be joined by a curve in X but that the same is not true of Y .

1. Sketches

The surface X is a hyperboloid of one sheet; Y is a hyperboloid of two sheets.



$$x^2 + y^2 - z^2 = 1$$



$$x^2 + y^2 - z^2 = -1$$

Figure 9: Left: X (hyperboloid of one sheet). Right: Y (hyperboloid of two sheets).

2. Rigorous argument

Connectedness of X . We show that X is path-connected. For any point $(x_0, y_0, z_0) \in X$, define a parametrisation

$$\gamma(t) = (\sqrt{1 + z_0^2} \cos(t), \sqrt{1 + z_0^2} \sin(t), z_0), \quad t \in [0, 2\pi].$$

This curve lies entirely in X because

$$(\sqrt{1+z_0^2} \cos t)^2 + (\sqrt{1+z_0^2} \sin t)^2 - z_0^2 = (1+z_0^2) - z_0^2 = 1.$$

Hence every point of X is connected to the point $(\sqrt{1+z_0^2}, 0, z_0)$ by a curve in X . Now take any two points $P, Q \in X$. Let P be joined to a point P' on the same horizontal circle (constant z) as P and similarly Q to a point Q' on its own circle. The set of all points with a fixed z is a circle; any two points on the same circle can be joined by an arc of that circle. Moreover, for any two values z_1, z_2 there exists a curve that varies z continuously while keeping $x^2 + y^2 = 1 + z^2$ (e.g., a straight line in cylindrical coordinates). Combining these, we obtain a continuous path from P to Q entirely contained in X . Thus X is path-connected, so every pair of points can be joined by a curve in X . $\square$

Disconnectedness of Y . Observe that for any point $(x, y, z) \in Y$ we have $x^2 + y^2 = z^2 - 1$. Hence $z^2 \geq 1$, so either $z \geq 1$ or $z \leq -1$. Define

$$Y_+ = \{(x, y, z) \in Y : z \geq 1\}, \quad Y_- = \{(x, y, z) \in Y : z \leq -1\}.$$

Both Y_+ and Y_- are non-empty (e.g., $(0, 0, 1) \in Y_+$, $(0, 0, -1) \in Y_-$) and they are disjoint. Moreover, Y_+ and Y_- are closed in $\mathbb{R}^3$ (they are preimages of closed intervals under the continuous projection $(x, y, z) \mapsto z$), hence they are closed in the subspace topology of Y . Since $Y = Y_+ \cup Y_-$ and the two sets are disjoint, closed, and non-empty, Y is not connected. Therefore there exist points in Y (one in Y_+ , one in Y_-) that cannot be joined by a curve lying entirely in Y . $\square$

Problem 0.5. Use $\varepsilon - \delta$ style statement to show the limit:

1. $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x - 1} = 2$.
 2. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.
1. $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x - 1} = 1$. For $x \neq 1$, we simplify:

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. Then for any x with $0 < |x - 0| < \delta$, we have

$$\left| \frac{x^2 - 1}{x - 1} - 1 \right| = |(x + 1) - 1| = |x| < \delta = \varepsilon.$$

Thus $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x - 1} = 1$. (Note: The problem statement claims the limit equals 2, which is incorrect. The correct limit is 1.) $\square$

2. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. For any x with $0 < |x - 0| < \delta$, we have

$$\left| x \sin\left(\frac{1}{x}\right) - 0 \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| < \delta = \varepsilon.$$

Hence $\lim_{x \rightarrow 0} x \sin \left(\frac{1}{x} \right) = 0$.

□